SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F78PA
Probability and Statistics A

Wednesday 13 December 2013
1.00 pm to 3.00 pm

Question	Marks
1	11
2	11
3	16
4	15
5	10
6	7
7	15
Total Marks	85

Candidates must attempt ALL SEVEN questions.
To receive full credit you must show your work and explain your answers.

Approved electronic calculators may be used

AMS Formula Sheet and Cambridge Statistical Tables are provided

1. For **each** of the two distribution functions given below:

   (a) Sketch the distribution function.

   (b) Determine whether the corresponding random variable is a discrete or continuous random variable (give reasons).

   (c) Determine the **image** of the corresponding random variable and its **expected value.**

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(I) Distribution function for random variable X:

$$F_X(x) = \begin{cases} 0; & x < -1 \\ 1=3; & -1 \leq x < 2; \\ 1; & 2 \leq x; \end{cases}$$

[5 marks]

(II) Distribution function for the random variable Y:

$$F_Y(y) = \begin{cases} 0; & y < 0; \\ y^2=2; & 0 \leq y < 1; \\ 1=2; & 1 \leq y < 2; \\ \frac{y-1}{2}; & 2 \leq y < 3; \\ 1; & 3 \leq y; \end{cases}$$

[6 marks]

[Total 11 marks]

2. Suppose that X is a continuous random variable with density function given by

$$f_X(x) = \begin{cases} Cx + \frac{1}{2} & 1 \leq x \leq 2; \\ 0 & \text{otherwise;} \end{cases}$$

where C is a constant.

(a) Show that $C = 1=3$. [2 marks]

(b) Determine $Var(X)$, the variance of X. [2 marks]

(c) Suppose that $Y = \ln X$. Determine ImY, the image of Y, and find $f_y(y)$, the density function for Y. [7 marks]

[Total 11 marks]

[PLEASE TURN OVER]

3. Suppose that X and Y are non-negative random variables with joint density

$$f_{X,Y}(x;y) = \begin{cases} \frac{1}{3}(x+y); & 0 \leq x \leq 1;\ 0 \leq y \leq 2; \\ 0; & \text{otherwise.} \end{cases}$$

(a) Determine $\mathbf{P}(2X > Y)$. [3 marks]

(b) Determine the marginal densities $f_X(x)$ and $f_Y(y)$ for X and Y, respectively. Are the random variables X and Y independent? (Give reasons for your answer.) [3 marks]

(c) Determine the conditional density function $f_{Y|X}(y|x)$ of Y given $X = x$, and hence, or otherwise, calculate:
(i) $\mathbf{P}\left(Y < 1 \mid X = \frac{1}{2}\right)$ and **(ii)** $\mathbf{E}\left(Y|X = \frac{1}{2}\right)$. [6 marks]

(d) Given that $\mathbf{E}(X) = 5{\cdot}9$ and $\mathbf{E}(Y) = 11{\cdot}9$, determine $Cov(X;Y)$ and comment on the association between the variables X and Y. [4 marks]

[Total 16 marks]

4. Suppose that X is a random variable with (marginal) density

$$f_X(x) = \begin{cases} e^{-x} & 0 < x < \infty; \\ 0 & \text{otherwise.} \end{cases}$$

and that Y is a random variable with conditional density, given $X = x$,

$$f_{Y|X}(y|x) = \begin{cases} \frac{2y}{x^2} & 0 < x < \infty; 0 < y < x; \\ 0 & \text{otherwise.} \end{cases}$$

(a) Sketch the region where $f_{Y|X}(y|x) > 0$ and show that for any $x > 0$,

$$\mathbf{E}(Y|X = x) = \frac{2x}{3}.$$

Hence, or otherwise, determine $\mathbf{E}(Y)$. [5 marks]

(b) Determine the joint density $f_{X,Y}(x;y)$ for the variables X and Y. [2 marks]

(c) Let $Z = Y{\cdot}X$ and show that $f_Z(z)$, the density for Z, is given by

$$f_Z(z) = \begin{cases} 2z & 0 < z < 1; \\ 0 & \text{otherwise.} \end{cases}$$

[8 marks]

[Total 15 marks]

[PLEASE TURN OVER]

5. Suppose that $X_1; X_2; \ldots; X_n$ are *independent* random variables with distribution functions $F_1(x); F_2(x); \ldots; F_n(x)$ respectively. Let $M = \min\{X_1; X_2; \cdots; X_n\}$.

   (a) Let $F_M(x)$ denote the distribution function for M. Show that for any $x \in \mathbb{R}$,

   $$F_M(x) = 1 - (1 - F_1(x))(1 - F_2(x)) \cdots (1 - F_n(x)).$$

   [*Hint:* Note that $\mathbf{P}(M > x) = \mathbf{P}(X_1 > x; X_2 > x; \ldots; X_n > x)$ for $x \in \mathbb{R}$.]

   [4 marks]

   (b) An electric torch requires two batteries to work. Suppose that X_1 is the lifetime (in years) of one of the batteries in the torch and X_2 is the lifetime (in years) of the other battery. In addition, suppose that X_1 and X_2 are *independent* and *identically distributed* with distribution function given by

   $$F_X(x) = \begin{cases} 1 - \frac{1}{x} & x \geq 1; \\ 0 & \text{otherwise.} \end{cases}$$

   Suppose that the two batteries are put in the torch at time $t = 0$ and let T denote the time at which the torch stops working (i.e. T is the time at which one of the batteries stops working).

   i) Use part (a) above to determine F_T, the distribution function of T. [2 marks]

   ii) Determine the *median* of the distribution function F_T. [2 marks]

   iii) *Given* that the torch is still working after 2 years, determine the *conditional* probability that the torch is still working after 3 years. [2 marks]

   [Total 10 marks]

6. (a) Suppose that $X_1; X_2; \ldots; X_n$ are i.i.d. random variables and let

   $$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

   denote the sample mean of $X_1; X_2; \ldots; X_n$. Show that the moment generating function, $M_{\bar{X}}(\ )$, for $\bar{X}$ is given by

   $$M_{\bar{X}}(\ ) = \left(M_X\left(\frac{\ }{n}\right)\right)^n$$

   where $M_X(\ )$ is the moment generating function for the variables X_i, $i = 1; 2; \ldots; n$.

   [4 marks]

   (b) Suppose that $X_1; X_2; \ldots; X_{20}$ are i.i.d. $\Gamma(2; 3)$ random variables. Determine the distribution of the sample mean $\bar{X}$. (Give reasons for your answer.) [3 marks]

   [Total 7 marks]

[PLEASE TURN OVER]

7. The number of accidents reported at a large factory in any given week has a $Poisson(2)$ distribution, independent of the number of accidents at the factory in any other weeks.

   Starting at $t = 0$, let X_1 denote the number of reported accidents in the first week of the year, let X_2 denote the number of reported accidents in the second week of the year, and, in general, for $n = 3,4,5,\ldots,52$, let X_n denote the number of reported accidents in the nth week of the year.

   (a) Let $Y = X_1 + X_2 + \cdots + X_{52}$ denote the total number of accidents reported in one year (i.e. 52 weeks) at the factory.

   State carefully the Central Limit Theorem and use it to approximate the probability that, in total, less than 90 accidents are reported in one year (52 weeks), i.e. approximate $\mathbf{P}(Y < 90)$. [7 marks]

   (b) The factory categorises an accident as 'serious' if an employee is injured in the accident. Based on accident data, the factory manager estimates that the probability that an accident is serious is $p = 1/4$, independent of all accidents.

   Given that $T_1, T_2, T_3, \ldots$, the times (in weeks) between accidents at the factory, are i.i.d. $Exp(2)$ random variables, determine the distribution of W, the time from the start of the year until the first *serious* accident occurs

   Hint: Find the moment generating function $M_W(\ )$ for W.

   [8 marks]

   [Total 15 marks]

[END OF PAPER]